

Public policies for improving food and nutrition security at different scales

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Abstract The challenge we face today is the achievement of zero hunger in the world by 2050. Enhancing food security not only enhances health and productivity, it contributes significantly to social wellbeing, economic development and national and global stability. Such achievements will call for effective public policies to influence the areas of demand and access through markets and supply. In relation to demand; many policies can target consumer demand for food, including instruments which influence employment and incomes, food preferences and consumer knowledge, health services and food safety. Policies which influence consumer access to food depend on functional value chains, equitable market environments, infrastructure and stabilisation policies – and above all creating enabling environments for business investment and engagement through incentives and regulations. The third group of policies analysed are those that influence producer supply by assisting in enhancing food production, including through rural infrastructure development, agricultural research and development, resource management, farm inputs and produce pricing. In this paper we consider a range of diverse policy approaches targeted at demand, access and supply that directly affect food and nutrition security with a view to better understanding which are most effective at addressing zero hunger. We conclude that the effectiveness of food security policies is determined by selecting the best bundle of policy instruments for the specific context and country and that tradeoffs between policy instruments should be well-understood, in order to achieve the right goals and avoid perverse outcomes.

Keywords Food security · Food policy · Demand · Supply · Markets · Nutrition

Introduction

Food is one of humankind's most basic needs – and according to Article 25 of the Universal Declaration of Human Rights of the United Nations it is a fundamental human right. Human development, economic development, peace and security all depend on nations being food secure. Policies that support sustained growth in food systems (from supply through to consumption) are critical drivers for inclusive economic development and for ensuring resilience. Brinkman and Hendrix (2011) provide an overview on how food insecurity – especially when caused by a rise in food prices – can be a threat and impact multiplier for violent conflict. While it might not be a direct cause and rarely the only cause, when combined with other factors, food insecurity could be the factor that determines whether and when violent conflicts will erupt. The 2008 global food price crisis exposed the vulnerability of the global food system and how quickly food insecurity can lead to significant civil unrest.

The definition of the right to food enshrined in the 1966 International Covenant on Economic, Social and Cultural Rights is the right to have regular, permanent and unrestricted access, either directly or by means of financial purchases, to quantitatively and qualitatively adequate and sufficient food corresponding to the cultural traditions of the people to which the consumer belongs, and which ensure a physical and mental, individual and collective, fulfilling and dignified life free of fear.

Food insecurity is a complex phenomenon of multiple factors and issues that cannot be solved by just one sector or institution. Evidence based policies, political commitment and social participation in policy processes are needed to

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create an enabling environment for resolving these issues and delivering food and nutrition security. Good public policy influences how people, sectors and institutions interact with each other and provide incentives to improve food security outcomes. An enabling environment has a range of dimensions that policy can influence, including food demand, production and access policies. Public policies are essential because market mechanisms alone cannot provide for all the resources needed for producing, storing and distributing food along the value chain, nor deliver the institutions and regulations required to underpin equitable and safe food systems.

Public policy is also needed to support communities to build resilience, or to bounce back following a negative shock. Effective public policies are especially important to economies consisting of smallholder farmers, who face increasingly volatile markets with price instability, climate change impacts and natural disasters (such as floods, droughts, diseases) and conflict. Price and income swings can significantly affect the poor. Public policy is also crucial to ensure the provision of public goods, such as ecosystem services or technology to facilitate changes in farming practices and to ensure a sustainable farming sector that meets societal demands. Public education policies are essential for educating households on nutritional requirements, sanitation practices and food safety.

This paper examines food policies and how they affect the main elements of the food system - demand, markets and supply. The second section discusses the evolving concept of food security and impacts of hunger and malnutrition. Adding a nutrition element to the food security definition ensures a focus not just on calorific value of food, but the micro-nutrient value also. Policies related to food demand are examined in the third section, including adequate income and assets, shifting food preferences as communities develop economically, and food price policies. The fourth section discusses the role of agricultural markets and how appropriate trade and market policies can help in improving food security through strengthening producer market participation. Policies related to food supply are discussed in the next section. These are framed around the concept of sustainable intensification built from adequate public research, infrastructure, inputs and resource use policies. Conclusions are provided in the final section.

An evolving concept of food security - hunger and nutrition

The concept of 'food security' has evolved over time with different dimensions and scales; from global and national levels to community to individual households. At the 1974 World Food Conference, held by FAO, food security was first defined as "...sustaining a steady expansion of food consumption and to offset fluctuations in production and prices". This reflects an earlier concept of food security where the

focus was solely on the supply side and overlooked the multiple variables that affect the demand and access sides. In time, the focus shifted from national to individual levels and a new definition emerged from the 1996 World Food Summit as food security "existing when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets dietary needs and food preferences for an active and healthy life" (FAO 2002; Van der Veen and Tagel 2011).

Food security pillars

The FAO food security definition is comprised of four inter-related pillars:

- *Availability* - ensuring adequacy of food supplies in terms of quantity, quality and variety of food,
- *Access* - optimising stability in the affordability and allocation of food, as well as the preferences of individuals and households,
- *Utilization* - ensuring food ingested is safe and is sufficient to meet physiological requirements of the individual. Elements include food safety, nutritional values, access to healthcare, sanitation and education,
- *Stability* - the ability to consistently obtain food over time. Food insecurity can be transitory (temporary decline in access – "hunger season"), or chronic (constant failure to access food).

Improving any one of the dimensions alone is necessary but not sufficient to improve food security. The FAO (FAO 2014) has developed a suite of indicators that provides a more comprehensive picture of food security across these four dimensions which can help in targeting priorities for food and nutrition policies.

The state of food insecurity

According to *The State of Food Insecurity in the World* report, one in nine people in the world go hungry every day (FAO 2014). Since the inception of the Millennium Development Goals (MDG), the world has committed to halving the proportion of hungry people between 1990 and 2015. Considerable progress has been made; with the number of hungry people reduced by 100 million over the last decade and by 209 million people since 1990–92. Yet despite this overall progress, marked differences exist across regions and 805 million people remain affected by chronic hunger, 165 million children are stunted (with levels as high as 58 % of the child population in some countries) and over 2 billion people are affected by micronutrient deficiency (IFPRI 2014; FAO 2014). The Sub-Saharan Africa region has the highest prevalence of undernourishment while the South

Asian region has the highest number of undernourished people. The regions of South-Eastern Asia, Latin America and the Caribbean have made significant progress in achieving their MDG hunger targets with many countries, such as Brazil, Cameroon, Ethiopia, Gabon, the Gambia, Iran, Kiribati, Malaysia, Mauritania, Mauritius, Mexico, the Philippines and Uruguay are on their way to eradicating hunger (FAO 2014).

Hunger perpetuates poverty by reducing people's ability to work, to learn and to lead prosperous lives and in recognition of this, there is renewed global momentum to end hunger and malnutrition. Malnutrition is defined as nutritional deficiency (under-nutrition) or nutritional excess (over-nutrition - Friel and Forde 2015).

Women and girls are the most vulnerable, with unique physiological requirements through pregnancy and breastfeeding that are frequently not understood and usually not catered for adequately in poor rural households. Additionally, they often carry heavy home and farm work burdens. Of this group, adolescent girls are especially at risk of low levels of iodine, iron, zinc and vitamin A - the big four micronutrients essential for a functioning immune system and cognitive development into adulthood.

The global research community has recognised the important role under-nutrition plays in creating and maintaining poverty traps, with maternal and child under-nutrition identified as the primary pathway by which poverty is transmitted from one generation to the next. Sound nutrition is instrumental in achieving physical, cognitive, social and economic development. The critical window of '1,000 days' covers the period from conception to 24 months in which developmental damage resulting from under-nutrition cannot be reversed or regained and will impact on the future quality of adult life across the lifespan. Multiple studies link childhood under-nutrition with reduced earning potential due to diminished adult worker productivity and for women, production of offspring with lower birth-weight allows the cycle of poverty to continue. The economic costs of micronutrient deficiencies are considerable, reducing gross domestic product (GDP) by 2–3 % in many low-income countries (IFPRI 2014), with one study documenting economic losses due to iron deficiency anaemia accounting for a median 4 % loss in GDP across the 10 developing countries examined (Horton and Ross 2003).

The state of global food security will be increasingly challenged as populations increase and food production is compromised by depleting natural resources (particularly land and water), climate change, pest and disease incursions and increasing energy costs. There is also an increasing demand for better quality food and more diverse diets as people move out of poverty (for example, the rising demand for meat and livestock products, especially in China and south-east Asia). This rising demand for more diverse and better quality food in

the diets, coupled with growing world population, has been estimated to drive a need to increase total global food production by 60–70 % by 2050 (FAO 2012).

Equity and gender policies

Globally, women provide 43 % of agricultural labour, with percentages as high as 60 % in some African countries (FAO 2011). They are however disadvantaged in various ways including lower access to resources, inputs and land, lack of voice in decision-making processes, including in agriculture research, and time poverty caused by the drudgery of agricultural and household tasks, amongst others. Providing women equal access to land, livestock, inputs such as seed and agricultural equipment, extension support and credit is essential. Women and men must be equal partners in improving global food security.

According to FAO, an increase in resources by women to the same level as men currently have could increase yields by 20–30%. This could raise total agricultural output in developing countries by 2.5–4%, which could in turn reduce the number of hungry people in the world by 12–17%.

In general, policy makers seek equitable outcomes, including gender concerns, although there are many social, institutional and political constraints to implementation (Crane and Gandhi 2009). Safety net policies, which are gaining popularity and directly improve purchasing entitlements and food security in stress conditions, are not analysed in detail in this paper. However, it is important to note that integrating safety net programs with agricultural development programs can be very effective, e.g., DFID programs in Ethiopia. There is a large literature on gender aspects of agricultural research and infrastructure development. However, subsidy programs and policies for improving gender outcomes is 'unfinished business' and require greater attention by policy makers. Research in Africa, Asia and Latin America has found that improvements in household food security and nutrition are associated with women's empowerment, such as access to income and increased role in household decisions on expenditure, as women tend to spend a significantly higher proportion of their income than men on food for the family.

Women's empowerment, through targeted agricultural interventions and safeguarding or strengthening the capacity of women to provide for the food security, health and nutrition of their families is the most effective pathway to food and nutrition security (World Bank 2013) and should be at the core of all efforts.

Policies for food demand

Sen (1981) introduced the idea of food security as a demand concern, where it is viewed in terms of entitlements,

representing the capacity to access food. In this regard, the ability of households or individuals to access food either through production, purchase or transfers is determined by a number of factors, policies and interventions. In this section we describe how food demand policies impact generally and how effective they have been in selected countries.

Income, growth and development

Having enough food in the market does not guarantee food security. The question of how much access households (or individuals) have to available food supplies depends heavily on their income or assets or other entitlements (Anderson 2014). A study in Kenya by Kirimi et al. (2013) showed that household commercialisation helped to reduce the risk of being chronically food poor and aided the transition out of food poverty. Income earned from other activities that might be related to employment can enhance the capacity of the household to access to food. This view is supported by Holden et al. (2004) who found that increased access to non-farm sources of income is likely to be good for household welfare, including food security. Employment in off-farm and cash crop activities provides farmers with an extra source of income and enables them to build resilience and reduce the risks of food shortage during periods of unexpected crop failures or between seasons when food supply is short. Self-sufficiency may provide smallholder farmers with a degree of food security, however, when farmers diversify their production to include high value crops or niche commodities and generate surpluses, they are able to transition into ‘enterprise’ farming practices that can increase their income. Diversified production provides better risk management as well as a more stable source of income and food supply. Governments can support household resilience through greater focus on agriculture and non agriculture sector development that generates greater employment opportunities.

Evidence suggests that it is not only access to the resources that matters, but also returns on those assets (Meinzen-Dick et al. 2011). The control of Newcastle disease in village poultry (owned and managed by women) in Tanzania has resulted in commercial opportunities and asset growth for women as flocks build. As poultry are traditionally seen as ‘women’s business’, an important outcome of this disease control policy is that women’s status has grown as they are able to keep ownership of the assets from their poultry and egg sales, including larger livestock assets (goats and cows) they have purchased with the profits. Asset ownership by women is often highly correlated with economic growth, poverty reduction and with reduction to vulnerability and risk at a household level (Njuki and Sangina 2013). The hypothesis that a lack of women’s access to household assets and resources leads to higher malnutrition rates in children, including in fisheries, has been described by many (Gehab et al. 2008), where a

key variable in determining household nutrition status is considered a woman’s relative bargaining power vis-a-vis men.

The poorest households in low-income countries spend more than half of their income on food (To and Grafton this volume; Regmi et al. 2001). Regmi et al. (2001) estimated that low-income countries spend a greater portion of their budget on food and are thus more impacted by income and food price changes than middle- and high-income countries.

Income growth drives much of the expansion in food demand. Non-OECD countries have increased their consumption of vegetable oils by 50 % (OECD, 2008). There is strong evidence that rapid economic growth (and high income) is transforming food demand in India and China (Gandhi and Zhou 2014). China and India, with their huge populations, have experienced high economic growth rates of 7 to 12 % in the last two decades. This has led to major changes in the levels and patterns of their food consumption and food buying behavior (Gandhi and Zhou 2014). Gandhi and Zhou (2014) show that food demand is undergoing a huge transformation and will undergo further change. They found that consumers are rapidly increasing their consumption of animal products, vegetables and fruits, and reducing their consumption of cereals. Their results show high income elasticities of demand for many food products such as animal products, processed foods, and also eating-out-of-home in both rural and urban areas. The 2012 Global Food Policy Report (IFPRI, 2012) reported that of the total global agricultural output, oil crops had increased from 6 % in 1970s to 8 % in 2009, with fruit and vegetables also increasing from 16 to 22 %. On the contrary, cereal production has decreased from 25 to 21 % over this same period.

Education and knowledge

Education is a key factor that influences the food security status of households. Education provides greater employment opportunities and increases household income. The important link between a woman’s level of schooling and her family’s nutritional status is well documented. Policies that remove gender barriers to learning and literacy and lead to the empowerment of women are critical for addressing food and nutrition insecurity. Education plays an important role in household awareness of nutritional and safety aspects of food rather than merely obtaining sufficient calorific quantity. How well households utilize food that is accessible to them will depend on their food nutrition, safety and hygiene knowledge and willingness to ensure a healthy and nutritious diet for all household members (Anderson 2014).

Food demand, particularly of specific commodities, is also determined by local production of the crop. Rural consumption of rice is directly correlated with production in the same province (FAO 2002). Policies should support dietary diversification and strengthen local food systems and incorporate

local and traditional knowledge on food preparation and agricultural diversification. Diet diversity can be achieved through a range of interventions, including consuming greater quantities and varieties of vegetables, fruits and animal-sourced food, fortification, biofortification, nutrition education and behavioural change. Better home and community food processing, preservation and storage and access to marketing facilities can also contribute to household food and nutrition security by alleviating seasonal shortages in food supply and stabilizing market prices. School Feeding Programs, when designed and implemented well, are good examples of policies that can deliver multiple outcomes. They aim to educate children on healthy eating as well as improve the diets of school children – often targeting specific nutritional gaps due to food supply at home – and generate new food demand. When meals are provided that are sourced from local food supply, they can also stimulate local production of nutritious foods that have widespread health and livelihood impacts across the community. One example of this is Brazil's Food Purchase Programme (Neves and Peduto 2010).

Food price stabilisation

Lower or subsidised food prices increase food demand (but reduce food supply, see section 5) and governments use this tool to make food more accessible to poor consumers. Subsidising food prices was a commonly used policy intervention when countries faced high food prices during the food price spike of 2006–08. Foreign exchange policies, affect the prices of imported and exported goods, which in turn affect both the volume and the structure of household food demand.

Restrictions on import and export of food can affect food price and its accessibility. In the 2008 food price spike, governments of some food exporting countries (including Argentina, India, Ukraine and Serbia) were concerned about the security of food supplies for their domestic consumers, and imposed restrictions on food export (Tangermann 2011). On the other hand, many food-importing countries reduced their import restrictions and a few even subsidized imports of their staple food. Both sets of actions have added to the international food price spikes and research shows that they have been much less successful in alleviating poverty (Anderson 2014) and enhancing food security.

In India (for example), governments applied highly interventionist policies (including a minimum support price (MSP) program) on food grains, especially rice and wheat, which supported and controlled domestic wholesale prices, large farm input subsidies, and consumer subsidy programs. The consumer subsidy scheme was implemented under the provisions of the Targeted Public Distribution System (TPDS) with the aim to more than double the distribution of highly subsidised food grains to cover approximately two thirds of the Indian population (Pursell 2014). Yu et al. (this volume)

argue that despite the high fiscal costs of the subsidised public distribution system which amounted to around 1 % of India's GDP in 2004–05, its food insecurity could not be reduced very much due to leakages and identification issues, and in the future improving access to food should be better dealt with by effective poverty alleviation measures.

A fundamental disconnect exists between what most countries say their food security strategies are and what policies they are actually pursuing. The disconnect is most manifest with rice policy, where high prices for rice farmers are implemented to reduce poverty, when in fact most of the poor and hungry in the region are net rice buyers and are negatively affected by high rice prices (Timmer 2014). The disconnect extends well beyond rice price policies, e.g., public research and development spending on agriculture is lagging badly, despite evidence that it has a very high payoff in raising agricultural productivity and reducing poverty (Suphannachart and Warr 2011; Ivanic and Martin 2010).

Policies for access through agricultural markets

This section recognises the importance of government policies to enable smallholder producers to participate in agri-business value chains, through equitable access to local, national or international markets. Well functioning markets can help both producers as well as consumers improve household food security and provide greater comparative advantages over subsistence production. Well designed market and trade policies can positively affect supply and demand and help stabilize commodity prices. Market prices provide signals to both exporting and importing countries about the opportunity cost of increasing production and/or consumption. Along with policies that advance infrastructure, policies that improve markets can promote efficiencies in postharvest value chains and input use (Pinstrup-Andersen 2009). Strong local agricultural markets and value chains where suppliers are connected to farmers and farmers to consumers can accelerate inclusive growth in the value of agricultural production and promote greater investment.

Effectiveness of markets

Well functioning global food markets can help increase incomes and improve food availability and affordability. With agricultural value chains becoming more complex, actions taken in one part of the world will no doubt affect smallholders or poor consumers in other parts of the world. The laws and regulations governing the different aspects of value chain development and trade agreements and institutions also directly impact market opportunity and productivity (GHI 2013). Global food markets can offer some local food producers access to larger markets, as well as to capital for

investment. At the aggregate level, they increase the global efficiency of food production by allowing regional specialisation in the production of the locally most appropriate foods. The high-value coffee and cocoa value chains are good examples, where most of the global production of these ubiquitous commodities is in a small number of regions by aggregated smallholder farmers. Because the expansion of food production and the growth of population occur at different rates in different geographic regions, global trade is necessary to balance supply and demand across regions (Godfray et al. 2010).

FAO (2007) argues that export-oriented economic activity is a major source of foreign exchange and income-earning opportunities along with providing nutritional and food security dimensions. The Global Panel on Agriculture and Food Systems (2014) recognize that trade policies on border movement of commodities can result in significant impacts on poor consumers and producers. Intra-regional trade is an important pathway to achieving food security in Africa, especially in times of crisis; however, the movement of food (including nutritious animal-based products) is hampered by inadequate phyto-sanitary policies and capacity at borders. Inadequate biosecurity implementation and capacity can also result in penalties or heavy costs to developing countries aiming to export commodities to global markets. Perishable food, such as vegetables and fresh animal products, are often more nutrient dense, yet if appropriate infrastructure is not available, these food products are not able to be moved, stored, or marketed (Global Panel on Agriculture and Food Systems 2014).

Government policies can help or hinder well functioning markets. Market friendly policies can reduce the gap between how much consumers pay and how much producers receive and make the country in general, and the agriculture sector in particular, better off. For example, In December 2010 China's National Development and Reform Commission implemented a major agricultural policy reform that impacted both producers and consumers by removing all toll roads nationwide and including in the road network 'easy access' for fresh agricultural product transportation (Zhong and Kong 2014). This decision was taken following soaring prices in 2008 and it helped to reduce the annual rate of vegetable inflation from a peak of 46 % in February to 22.7 % in March and by November that year it was -2.1 % (Chiang, L. & Edwards, N 2012). Anecdotal evidence suggests that this decision increased agricultural production and household income. Chen and He (2015) examined the effects of road tolls on economic growth in China and found that a 1 % rise in road tolls led to 0.005 % and 0.03 % reduction in GDP growth and GDP per capita, respectively.

Access to local markets

Access of smallholder farmers to local markets where they can participate in key services, including market information,

farmer group marketing, contract farming, enforcement of contract law, financing of small traders and local storage facilities plays a significant role in improving food and nutritional security. Smallholder farmers often are forced to sell at harvest when they are cash flow poor and have limited access to real credit. A recent study by Kirimi et al. (2013) identified factors that influence household food security for Kenyan rural smallholder households and, in particular, determined whether household participation in agriculture markets affects food security. They found a strong association between household commercialisation and reduced risk of being chronically food poor.

Local markets can bridge the gap between supply and domestic urban markets and trade and help both producers and consumers. The Public Distribution System (PDS) in India (for example) is said to have existed from before independence in India, and was initially intended to protect consumers from food shortages and producers from price fluctuations. As in India and other countries, the Food Products Procurement Programme was introduced in Brazil to ensure a market and reasonable price for products from smallholder farmers. This program involved a range of measures including: direct procurement of products at harvest for maintaining local food security stocks; advanced procurement of products at planting time; local procurement by local governments to be used in school feeding programmes; and a programme supporting milk production and consumption, benefiting producers with limited production and bargaining power (Nehring and McKay 2013).

Policies to insulate local markets from global markets

Some countries have taken protective measures to assist their producers while others took regulatory measures to insulate the domestic price and protect their consumers. Demeke et al. (2009) counted 36 countries that used some form of border intervention to stabilize their domestic food prices during the 2007–08 crisis.

Anderson (2014) argues that for a small country which cannot influence its terms of trade, any subsidy or restriction on its exports or imports would lower national economic welfare by foregoing some of the gains from production specialisation and exchange that trade openness provides.

FAO (2007) concludes that trade liberalisation is not a sufficient condition for promoting economic development, and calls for a more comprehensive and integrated policy, in which trade policy plays a central role in conjunction with other sectoral, national and regional policies. Expanded trade can provide insurance against regional shocks in production such as conflict, epidemics, droughts, or floods - shocks that are likely to increase in frequency as climate change occurs. Conversely, a highly connected food system may lead to the more widespread propagation of economic perturbations, as in

the recent banking crisis, thus affecting more people. There is an urgent need for a better understanding of the effects of globalisation on the full food system and its externalities (Godfray et al. 2010). Price stabilisation, increasing supplies available in local markets and providing safety nets to poor consumers, helps both producers and consumers. Also crucial are better infrastructure, governance and institutions. According to Timmer (2014), food price stability is a public good, not a market good. The cost of providing stability as a public sector activity can be very high, especially if public food agencies are poorly managed and become highly corrupt. Political action and public response from governments are essential to stabilize food prices.

According to GHI (2013), lowering tariff and non-tariff barriers to trade will remain a priority, as will approaching the rules and regulations around agriculture in a holistic, market-driven way. Positive developments are taking place at the intersection of trade, agriculture, and food security, but trade needs to be further integrated and better used as a tool for market development and productivity enhancement. In order to open markets effectively and to the benefit of all, innovation from both the public and private sectors will be increasingly important, as will creative and practical ways to combine the two (GHI 2013).

Policies for supply

This section describes the role and effectiveness of government policies on global food supply which must increase by 60–70 % by 2050 if projected demand is to be met (FAO 2012). The challenge is massive because of pressure on natural resources and is best viewed in the context of sustainable intensification (Godfray et al. 2013). At the household level, five main pathways for improved livelihoods and food security have been identified: intensification of existing production mixes; diversification of farm enterprises; expansion of the farm operation; increased off-farm income; and exit from agriculture (Dixon et al. 2001). At the aggregate global level, it is expected that most of the required increase in production would be derived from more ‘eco-efficient’ use of agricultural resources and inputs (Keating et al. 2013). Many of the examples of agricultural intensification during the last few decades illustrate the importance of public investment choices and policy as drivers of increased productivity (Spielman and Pandya-Lorch 2009; Fuglie et al. 2012; Fischer et al. 2014).

These examples show the complementarities and trade-offs between productivity and supply, on the one hand, and environmental equity and other social and political goals on the other hand (Lee and Barrett 2001). Potentially, the range of supply-side policy choices is very wide. They include, for example, policies aimed at increasing resilience to, or recovery from climate change impacts and natural disasters, e.g., the

Indus Valley floods in Pakistan, or drought in Ethiopia. Effective drought mitigation policies have been implemented in many countries, e.g., India and Australia; and crop and livestock insurance schemes are being piloted, with some degree of success, in many developing countries. However, the following sub-sections focus on four main supply-side investment and policy choices, viz, rural infrastructure, agricultural research and development (R&D), institutions for agricultural resource management, and input and produce pricing.

Rural infrastructure

The FAO (2012) emphasizes the importance of rural infrastructure for agricultural intensification and food supply, notably transport, energy, irrigation and market infrastructure – historically, infrastructure has been a driver of increased production, second only to agricultural R&D (discussed below).

Fan and Chan-Kang (2008) identify public investment in infrastructure notably secondary roads as a major determinant of agricultural and rural development and poverty reduction in China. In the Lao PDR, Warr (2005) attributes about 13% of the decline in rural poverty during 1997–2003 to improved road access. Similarly, in India, Fan et al. (2000) estimates high returns to rural road development in both irrigated and rainfed farming systems, although the returns in rainfed areas varied substantially across different farming systems. In many countries public investment in irrigation facilities has stimulated production and reduced rural food insecurity, e.g., China, India and Ethiopia. In Asia, rural subsidised electrification has increased irrigated water availability. Nevertheless, there are trade-offs: free water provision in northern India has led to ground water depletion. Recently, effective energy and water pricing mechanisms have been introduced into Gujarat which show promise for other developing regions.

The spatial targeting of public infrastructure development is critical: many African governments have adopted policies for economic corridor development, e.g., the Beira development corridor in northern Mozambique (Weng et al. 2013). However, the impacts of corridor development policies have been limited to date. Policies which promote private domestic investment in agricultural production and supply chain infrastructure are important, as farmers and agribusiness investment exceeds both domestic public and foreign direct investment in the sector.

Agricultural research and development, and extension

Agricultural innovation is the key driver of agricultural productivity (World Bank 2007), the most widely-cited example being the Green Revolution with dramatic improvements in food crop varieties and agronomic practices. Following the Green Revolution, complacency led to declining public investment in agricultural R&D, and ultimately to the food

crises of the 1990s and more recently 2008. Even so, Warr (2014) showed that Indonesian agricultural research during 1975–2005 accounted for a large proportion of the productivity growth and rural poverty reduction.

Ivanic and Martin (2010) found significant reduction in output prices (23.4 %) and poverty (4.6 percentage points from 33.8 %) in developing countries with an increase in investments in R&D and agricultural total factor productivity of 1 % per year, highlighting the fundamental importance of investment in agricultural R&D.

National agricultural research and extension systems, largely domestically funded, are complemented by the international CGIAR system, largely publicly funded. Relative investment in R&D can be measured by research intensity (agricultural research expenditure compared with agricultural GDP) and averages 3 % in developed countries compared with 0.5 % in developing countries (Beintema et al. 2012; Fischer et al. 2014). A key and urgent policy action is a massive increase in investment in agricultural R&D (Alston et al. 2009; Fischer et al. 2014), especially on research for increased sustainable productivity.

Public policies, which reinforce the trend towards greater private investment in agricultural R&D, are also required (Alston et al. 2009). In the USA, private investment in agricultural research now exceeds public funding, and is growing rapidly in many middle-income developing countries, e.g., Brazil and China, although lagging in India and Ethiopia. Similarly, yields and production have increased more rapidly for crops with substantial private sector investment (Dixon et al. 2009). Thus, incentives for private sector investment in food crop research to increase supply are needed and should be complemented by appropriate regulations (e.g., intellectual property).

Institutions for agricultural resource management

Improved conservation and management of natural resources, in particular, land and water resources, will contribute to a medium-long term supply sustainability. We contend that policy makers should focus on the reform of local and national resource management institutions. Changes to land tenure arrangements have underpinned major technical change and productivity increases in many countries, e.g., European countries, China, Ethiopia. The identification of effective policy instruments to improve land tenure is widely recognised as a complex and enduring challenge for policy makers in developing countries. HLPE (2011) recognises the diversity of institutional situations, and recommends a ‘multi-form’ approach to land tenure, mixing legal and administrative modalities to suit local circumstances. Given the diversity of often-conflicting stakeholder interests, extensive consultation is required following the principles of free prior and informed consent (FPIC), as well as the customary rules and

decision-making structures of local communities. Overall, policy makers should invest in local and national institutional capacities for land management, as well as strengthening community land rights registration in a consultative fashion.

Water management institutions are another critical policy challenge (see also Williams et al. this volume). Theoretically, the pricing of irrigation water will result in more efficient utilisation, but its introduction is often politically difficult especially without clear definition of property rights (Cruse and Gandhi 2009). Beyond the strengthening of institutions for irrigation management, improved institutions for the management of ‘green water’ (soil moisture in rainfed crops, pasture and range) is an important, largely ignored, area requiring policy attention. Gaiha and Kulkarni (2005) found that Indian policy interventions such as water harvesting schemes significantly contributed to a higher likelihood of household food security. In both cases institutions play a dominant role and decentralisation policies have had some degree of success, particularly in the transfer of irrigation management to communities or water user associations (Cruse and Gandhi 2009).

Farm input and produce pricing

Farm input subsidies are very common in Europe, North America and many developing countries, targeting seeds, fertilisers, credit, machinery and energy. During the past two decades Indian and Chinese policy makers successfully liberalised their seed sectors, but such policy lessons are only being slowly adopted in many less developed countries, e.g., Ethiopia. Fertiliser subsidies have been a major expenditure in many countries, e.g., India, even though such subsidies were difficult to finance when international oil prices rose to exceptional levels during 2006–7. In a well-known case, Malawi provided heavily subsidised seed and fertiliser input packages which increased maize production dramatically in good seasons – but have proved to be financially unsustainable.

Credit schemes for farmers have also received widespread subsidy support in many developing countries, but many publicly funded schemes were relatively ineffective in expanding food supply. However, group credit systems, e.g., those developed in Bangladesh by the Grameen Bank have been widely adapted and adopted with significant success in many countries. One key policy lesson is the need for integration of rural credit into the wider banking and financial systems, including mobile money initiatives such as M-Pesa in Kenya. As with the seed sector, most policy makers now support the delivery of financial services by the private sector.

The growing integration of rural, and urban labour markets in developing countries has led to increased wage rates for farm labourers and local shortages, for which subsidised mechanisation is one public sector response (other responses are changed practices and enterprise mixes, e.g., China, as well as abandonment of land, e.g., Nepal). Given that many

publicly-funded mechanisation schemes foundered, policy makers in developing countries now need to consider providing incentives for developing viable agribusiness solutions for the commercial delivery of small-scale mechanisation.

Although stabilisation schemes for the prices of farm produce are less common now, produce prices have been supported directly in many developed countries, notably Europe and the USA, as well as developing countries, e.g., India, for welfare reasons as well as an incentive for farmers to increase food production. While supply elasticities vary widely depending, *inter alia*, on farming systems, available technologies and market linkages, Wilson (2005) found that a one per cent increase in the real minimum producer support price for wheat in India would increase real food grain production per rural worker by 1.3% in the long run.

Conclusions

Feeding the projected global population of more than 9 billion and achieving zero hunger by 2050 at global, national and household levels is a major challenge for humankind. Clearly, public policy plays a critical role in addressing food and nutrition security. While policies to promote household food and nutrition security are principally a matter for developing countries, global food security involves both developed and developing countries.

Given that food and nutrition security involves multiple sectors, policies need to be coordinated across various sectors to foster an 'enabling environment' that provides the right incentives for all sectors to improve their impact on food security. Moreover, policy impacts can be counter-intuitive, not least because many food producers are net consumers and policy effects on food supply and demand can be opposite and perverse. Thus, it is particularly important that decision-making on food and nutritional security should be evidence-based and incorporate monitoring and research findings. The possible positive and negative tradeoffs between policy instruments should be well-understood as they are being designed, in order to achieve the right goals and avoid perverse outcomes. The effectiveness of food security policies is also significantly influenced by the specific context and country.

We have identified the most effective policy instruments to influence consumer demand as being those that facilitate greater on-farm and off-farm employment opportunities and greater agricultural and non-agricultural growth and development. In addition, we would recommend policies that promote education and support for local production, and appropriate agriculture protection measures which don't discourage investment and innovation.

Markets link demand and supply and govern access to food and nutrients. Effective markets fully depend on agri-business investment which calls for commercial incentives and

efficient regulations. Some of the most important policy areas we believe should be considered are those that facilitate well functioning local markets where suppliers are connected to smallholder farmers and smallholder farmers to consumers, greater infrastructure development and good governance and regulations (such as those concerned with phyto-sanitary and sanitary matters). Appropriate policies to insulate local markets from global price shocks resulting in reasonable stability in market price for smallholder farmers and consumers are warranted in the right context. We note the lack of harmonisation between supply policies and demand policies as an area that requires further attention for ensuring food security.

Policies to foster increased production and productivity of agriculture in a sustainable, inclusive and equitable manner are essential components of the bundle of public policies to improve food and nutrition security. Wider more equitable access to inputs (e.g., seed, water, fertilisers and credit) by smallholder farmers, including women farmers, would deliver a substantial dividend to this end, and contribute to several global sustainable development goals. In addition, sustainable food and nutrient supply will require policy makers to complement access to inputs through policies that improve resource management as well as allocate greater investments to agricultural R&D, extension and rural infrastructure.

The selection of an effective bundle of demand-side, market access and supply-side policy instruments which foster equitable food security outcomes, including for women and marginalised populations, remains a major challenge for policy makers.

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